

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14 (Old Course)

Paper No. CHB-401: Inorganic Chemistry-II and Selected Topics

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Inorganic Chemistry-II – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) How are pure superoxo and peroxo salts prepared in liquid ammonia?
- (b) Why are Cu^{2+} complexes colored, whereas Cu^+ complexes are colorless?
- (c) Why are ammine complexes of lanthanide ions not formed in aqueous solutions?
- (d) Why is NH_3 a stronger Lewis base than NF_3 ?
- (e) Define double salts and coordination compounds with examples.

Question 2.

- (a) State the Brønsted-Lowry definition of acids and bases. Illustrate Lewis acids and bases with equations. [6]
- (b) In terms of HSAB principles, which end of the SCN^- (thiocyanate) ion would you expect to coordinate to Co^{2+} and Cr^{3+} ? Explain. [6]

Question 3.

- (a) Write the IUPAC names of the following coordination compounds: [6]
 - (i) $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]^{2+}$
 - (ii) $[\text{Cr}(\text{en})_3]\text{Cl}_3$
 - (iii) $\text{K}_2[\text{Zn}(\text{NCS})_4]$
- (b) Depict the structures and write the names of the isomers of $[\text{Co}(\text{en})_2\text{Cl}_2]^+$. [6]

Question 4.

- (a) Discuss briefly three classes of aprotic solvents with examples. [6]
- (b) Write outer electronic configurations of Sm^{2+} and Tb^{4+} and indicate the number of unpaired electrons in each case. [6]
- (c) Why do lanthanide ions show a weaker tendency towards complex formation as compared to transition metal ions? What is characteristic about their coordination numbers? Explain. [6]

SECTION B

(Selected Topics in Chemistry – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) What is the basic difference between a fuel cell and a battery?
- (b) What is biomass? How can it be used as a renewable source of energy?
- (c) Discuss the cathodic protection method for the prevention of metallic corrosion.
- (d) Give a brief account of photoisomerization involving rotation about a C=C double bond.
- (e) Mention some of the limitations of the applicability of thermodynamics in biological systems.

Question 6.

- (a) Describe the construction and working of the H₂-O₂ fuel cell. Mention its components, cell reactions involved, and maximum theoretical e.m.f. obtainable. [8]
- (b) Discuss various methods used for the disposal and recycling of polymer wastes. [6]

Question 7.

- (a) Name different types of metallic corrosion, illustrating them in brief. [6]
- (b) Describe the weight-loss method for the determination of corrosion rates. [6]
- (c) What is rust? Write down the mechanism of rust formation on iron. [2]

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